

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25.parquet

BiGRU Multi-Step Training Report

Report Generated: 2025-05-07 08:57:21

Training Start: 2025-05-06 20:30:52

Training End: 2025-05-06 20:38:34

Training Duration: 7.70 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 2.6024

MSE: 9.8975

RMSE: 3.1460

R2 Score: -0.7304

Multi-Step Forecast:

Forecast Steps: 20

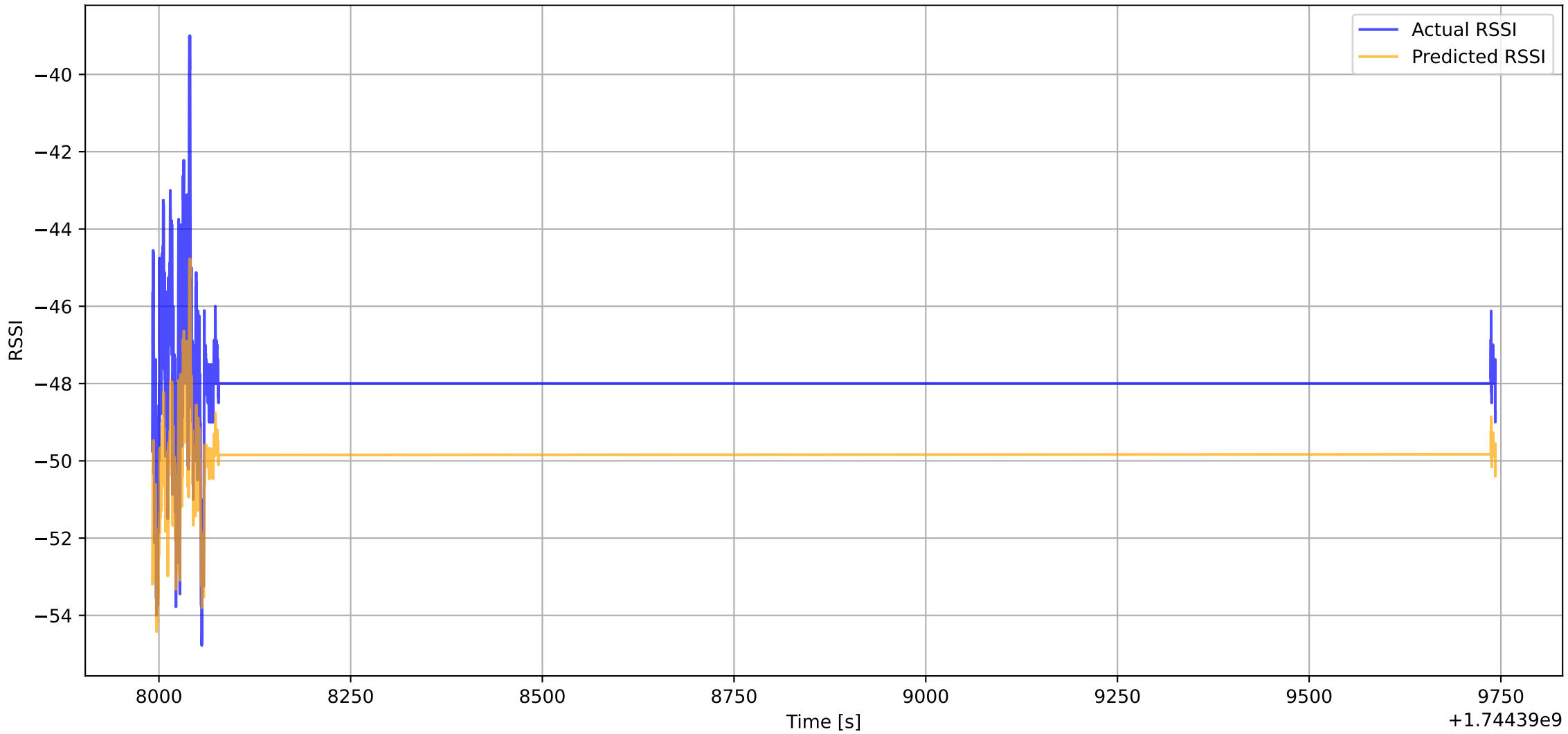
MAE: 0.0907

MSE: 0.0103

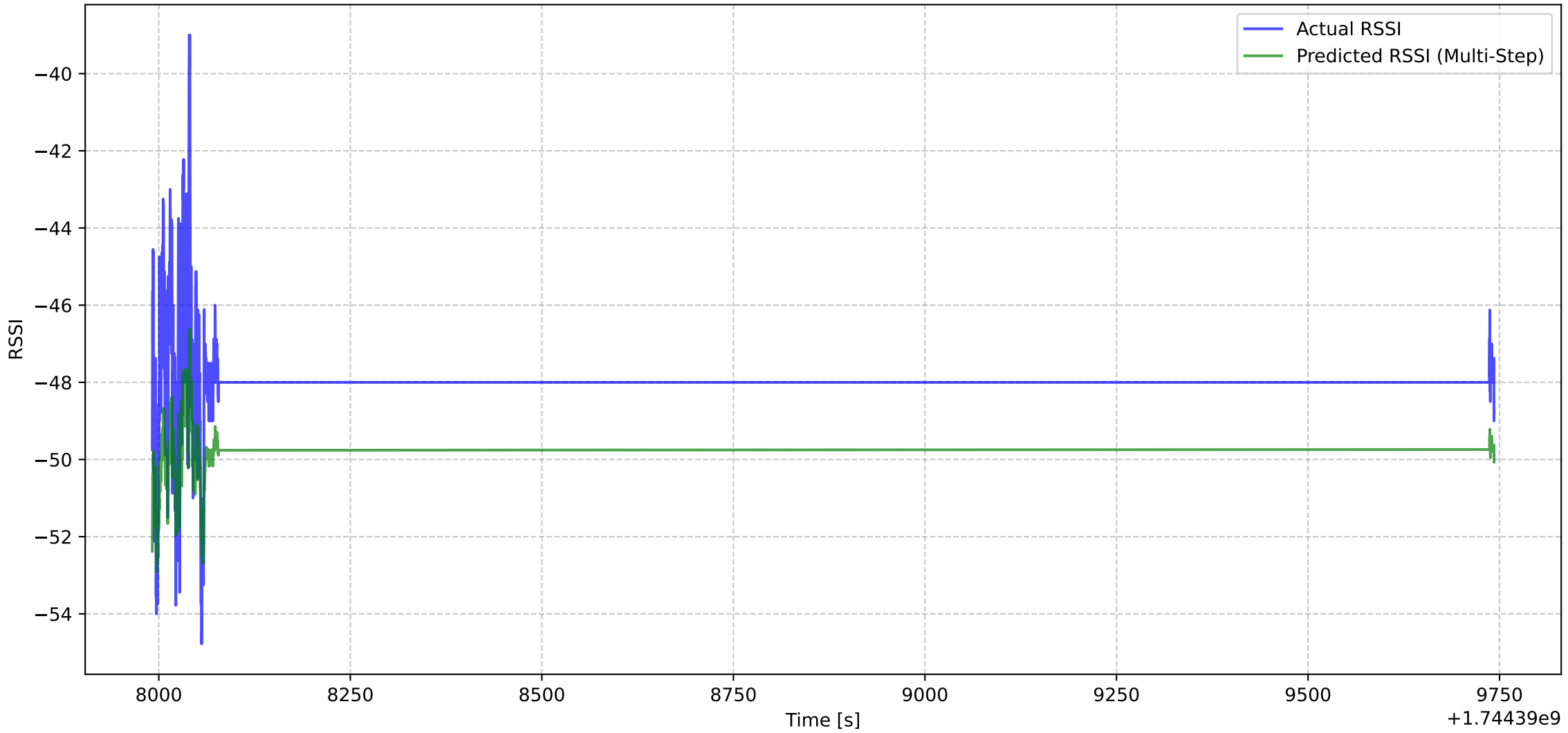
RMSE: 0.1014

R2 Score: -0.2765

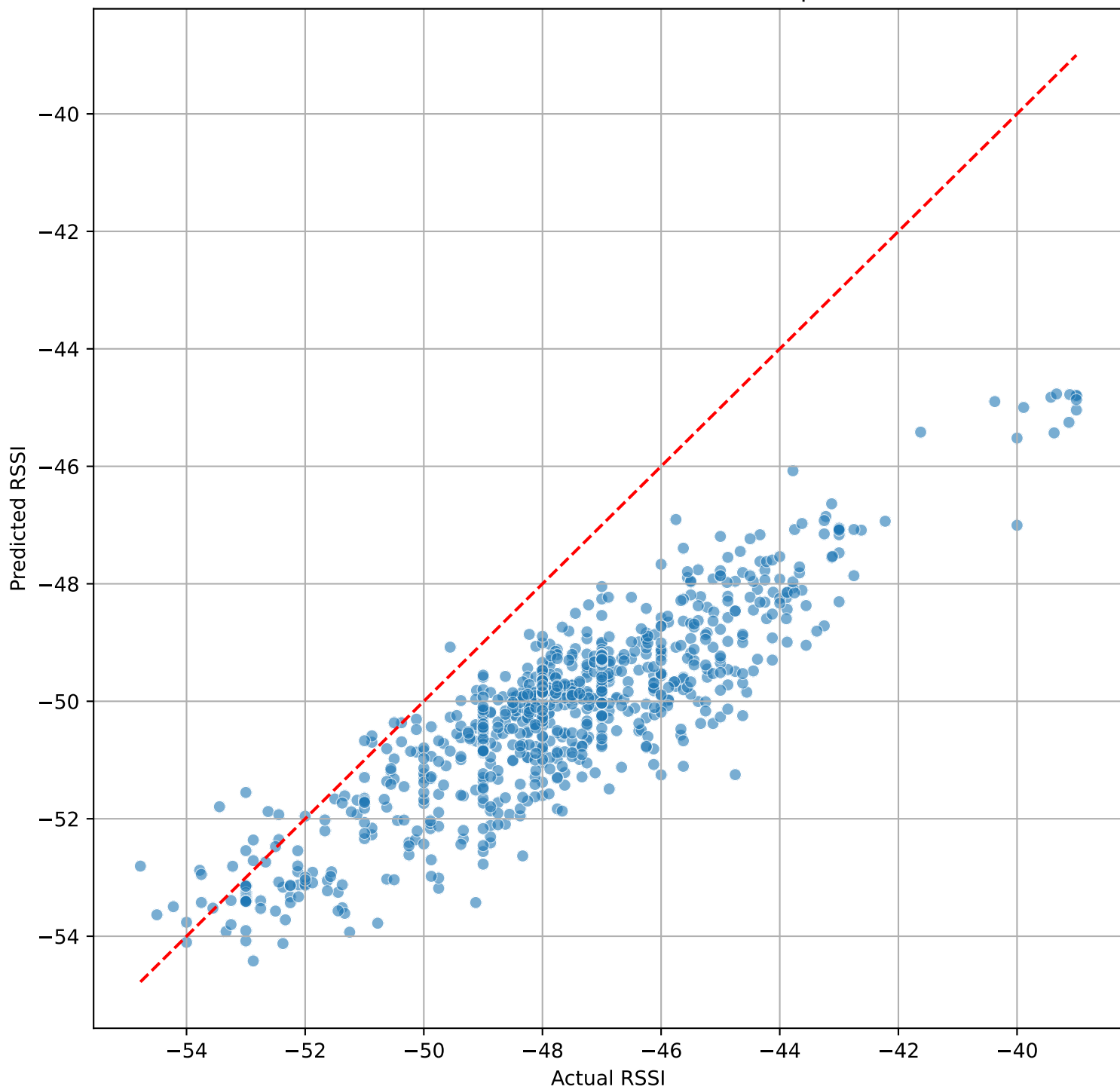
Actual vs Predicted RSSI (First Step)



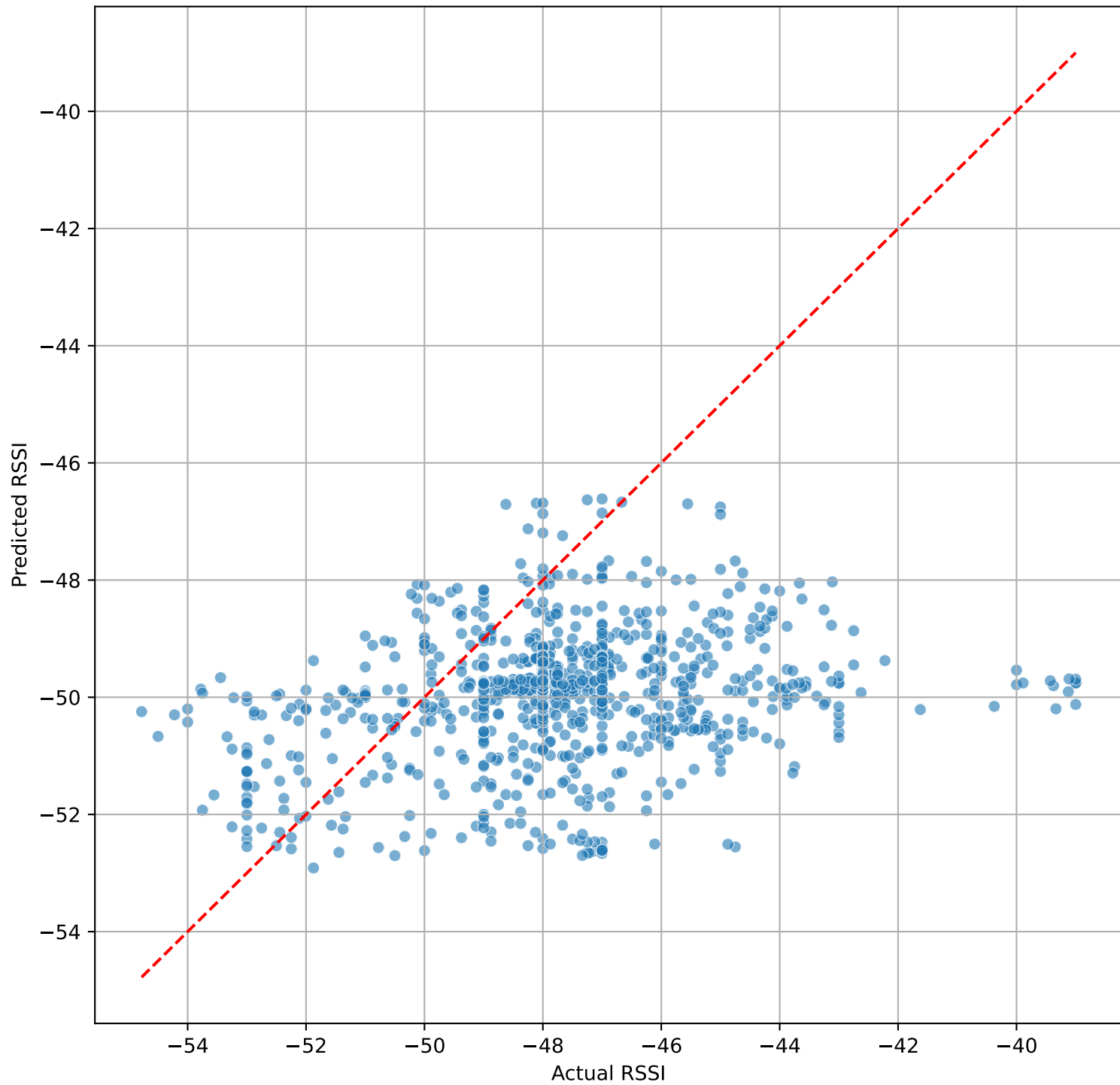
Actual vs Predicted RSSI (Multi-Step (step 20))



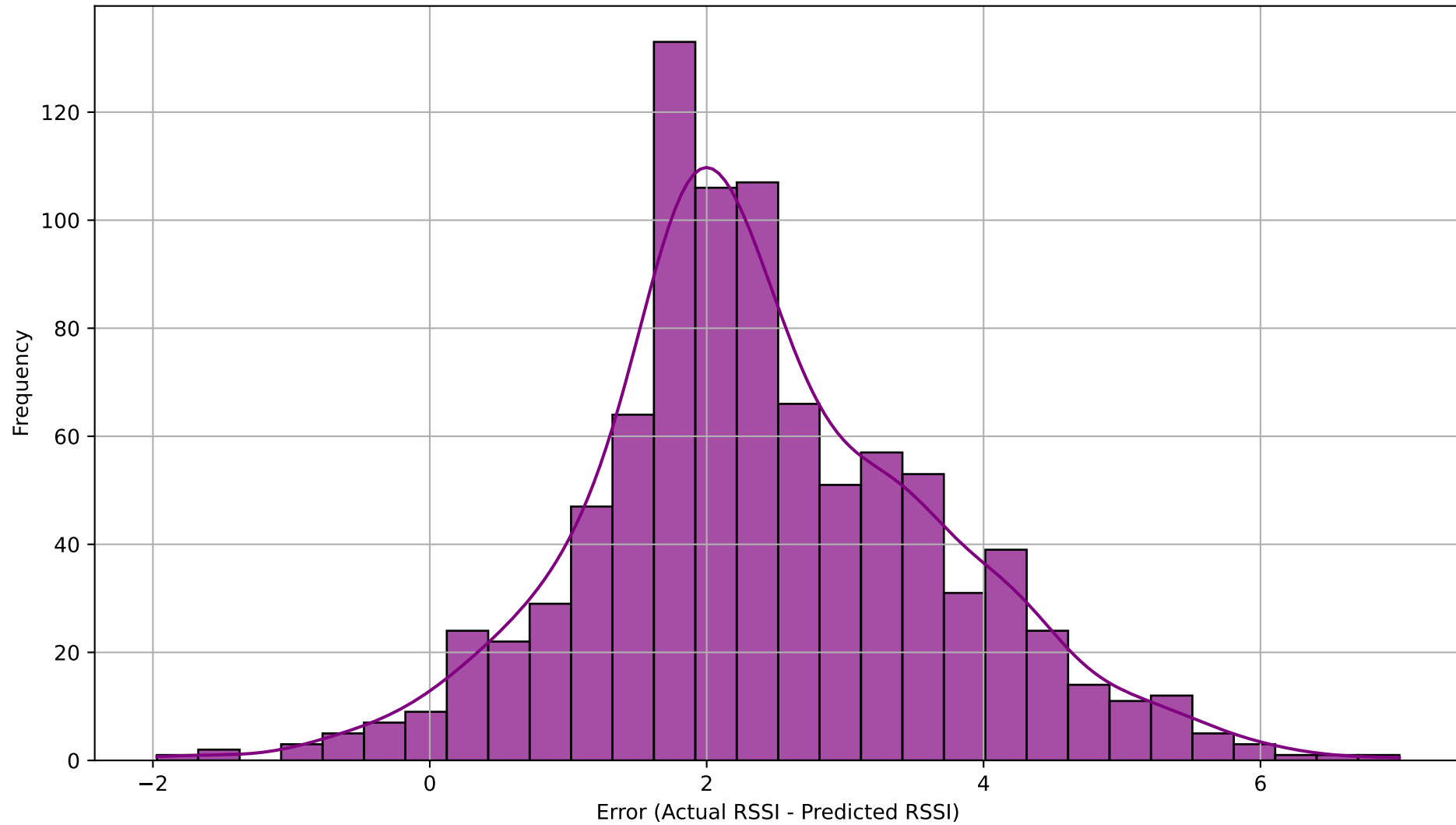
Actual vs Predicted RSSI (First Step)



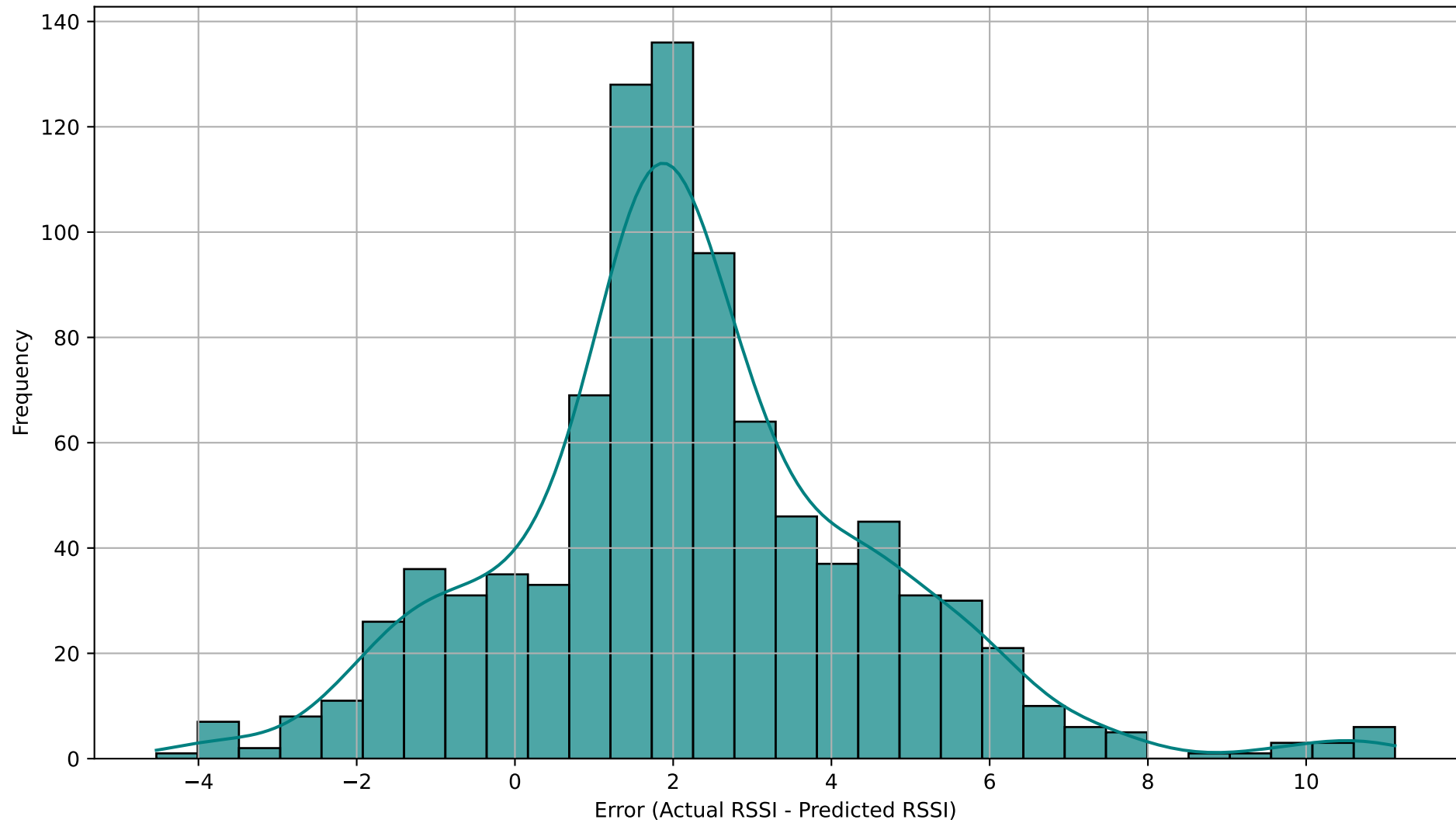
Actual vs Predicted RSSI (Last Step - step 20)



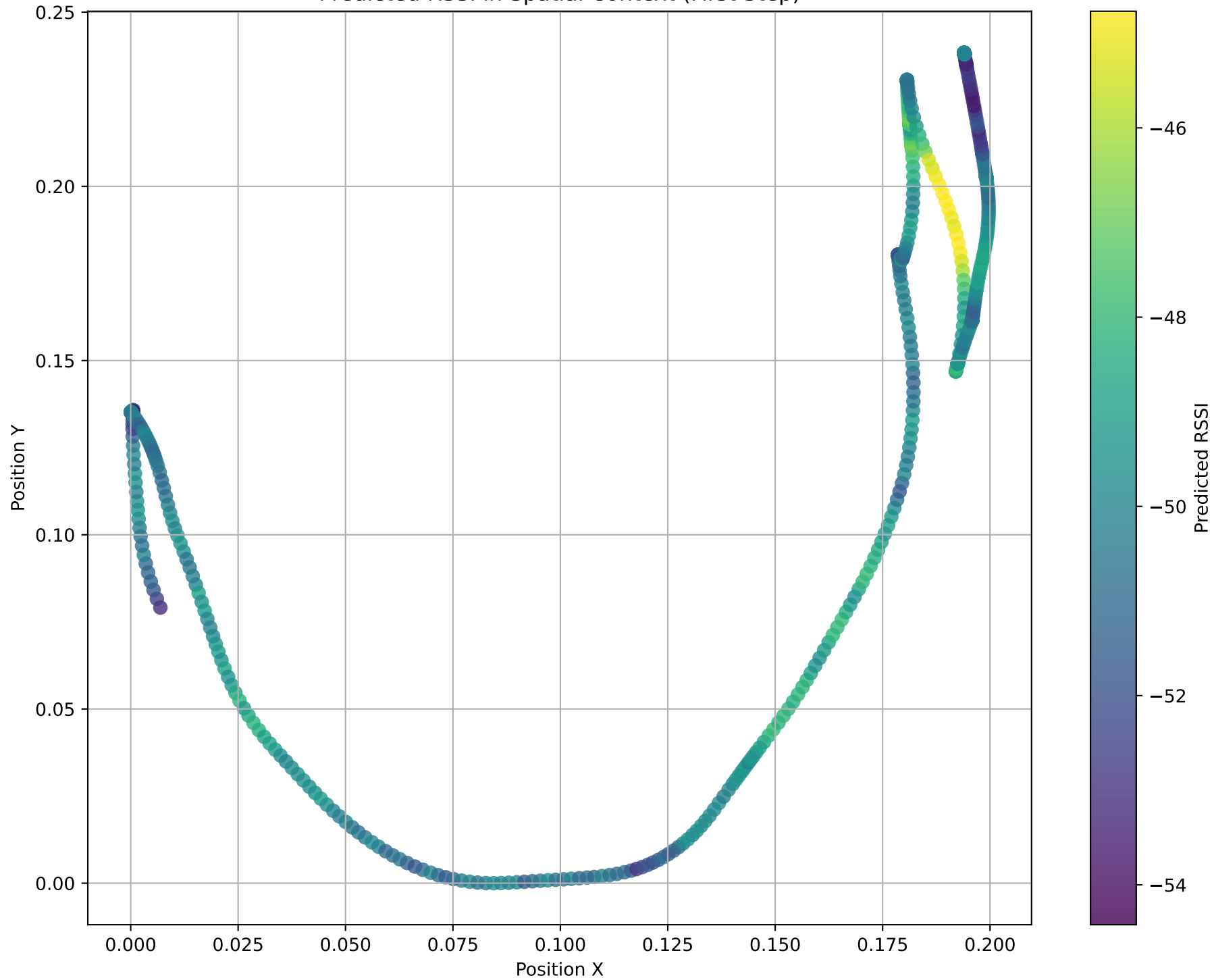
Distribution of Prediction Errors (First Step)



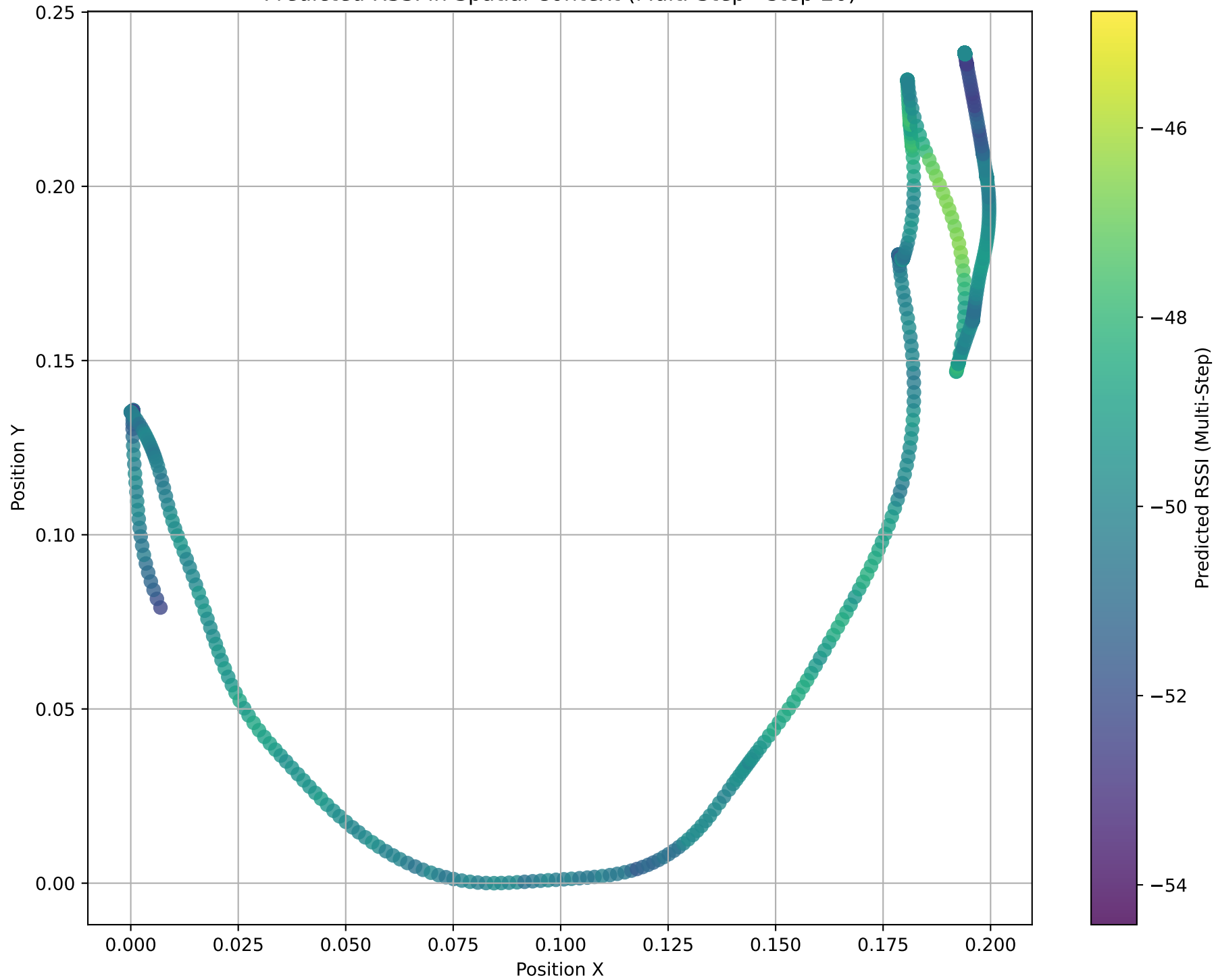
Distribution of Prediction Errors (Last Step - step 20)



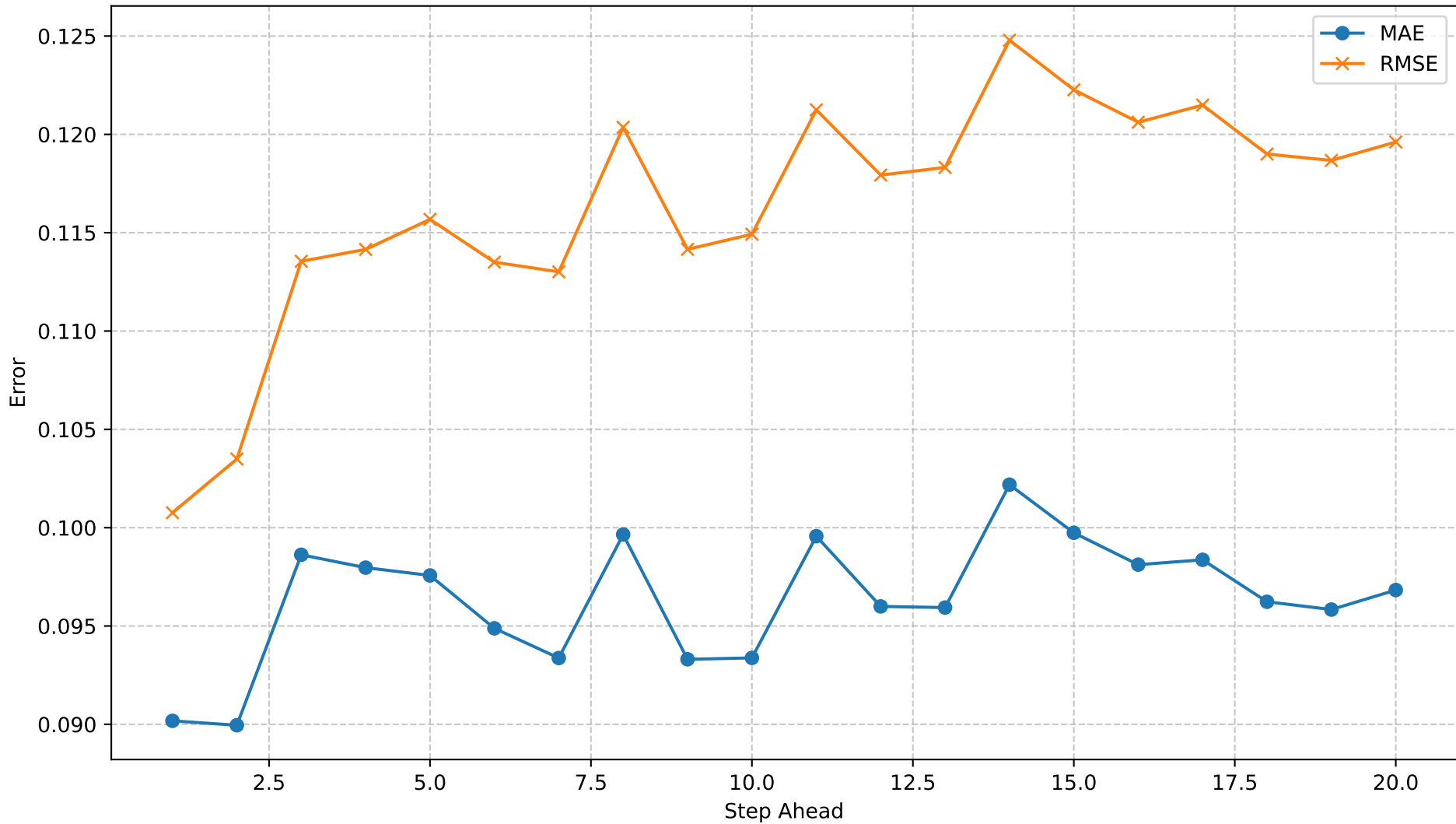
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25.parquet

Modelo: BiGRU

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.09 / 0.10
- Mediana : 0.10 / 0.12
- 20º passo : 0.10 / 0.12

Crescimento do erro: 7.4% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 14 (0.10), 15 (0.10), 8 (0.10)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.